

Detection of Criminal Communication Networks in the Enron Accounting Fraud Using Topic Modelling and Graph Neural Networks

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ABSTRACT

The Enron accounting fraud remains one of the most significant corporate scandals in history, involving complex communication patterns among employees that contributed to financial misconduct. Detecting such criminal communication networks is a challenging task due to the large volume of emails, hidden relationships among employees, and the highly imbalanced nature of fraud-related data. Traditional machine learning approaches often fail to effectively capture both the semantic content of communications and the structural relationships within organizational networks.

This project presents a graph-based framework for detecting criminal communication networks in the Enron email corpus using Topic Modelling and Graph Neural Networks (GNNs). The methodology begins with extensive preprocessing of the Enron email dataset, including data cleaning, email normalization, duplicate removal, text preprocessing, and employee identity mapping. BERTopic is employed to discover latent communication themes within email content, enabling the extraction of topic-based behavioral features. These topic representations are combined with communication statistics, graph centrality measures, Node2Vec embeddings, and contextual text embeddings to create a comprehensive feature set for each employee.

A communication graph is constructed where nodes represent employees and edges represent email interactions. Multiple Graph Neural Network architectures, including Graph Convolutional Networks (GCN), GraphSAGE, Graph Attention Networks (GAT), and Topology Adaptive Graph Convolutional Networks (TAGCN), are evaluated for criminal node classification. To further analyze network behavior, Louvain Community Detection and Spectral Clustering are applied to identify suspicious employee groups and examine the distribution of criminal nodes across communication communities.

Experimental results demonstrate that TAGCN achieved the best overall performance, successfully leveraging both node features and graph topology for criminal node detection. Community analysis revealed the presence of highly concentrated suspicious groups containing a significant proportion of known criminal employees. The integration of topic modelling, graph representation learning, and network analysis provides valuable insights into organizational communication patterns and enables effective identification of potentially fraudulent actors within large-scale email networks.

The proposed framework demonstrates the effectiveness of combining Natural Language Processing, Topic Modelling, Graph Analytics, and Graph Neural Networks for fraud investigation and criminal network detection in corporate communication systems.

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Chapter 1: Introduction

In today's digital world, organizations generate massive volumes of electronic communication through emails, messages, and collaborative platforms. These communication records contain valuable information about interactions between individuals and can provide critical insights into organizational behavior. However, they can also conceal fraudulent activities, insider threats, and criminal collaborations that are difficult to detect through traditional analysis techniques.

The Enron Corporation scandal, one of the largest corporate fraud cases in history, highlighted how internal communication networks can play a significant role in financial misconduct. The release of the Enron email corpus has provided researchers with a unique opportunity to study real-world communication patterns and develop intelligent methods for identifying suspicious behavior within large-scale organizational networks.

Traditional fraud detection approaches primarily focus on individual records or content-based analysis and often fail to capture the complex relationships that exist between employees. Emails contain not only textual information but also valuable structural information regarding who communicates with whom, how frequently they interact, and the role individuals play within the communication network. Therefore, an effective detection framework must analyze both the semantic content of communications and the underlying network structure.

Recent advancements in Natural Language Processing (NLP) and Graph Neural Networks (GNNs) have enabled more sophisticated analysis of large communication datasets. Topic Modeling techniques can uncover hidden themes and discussion patterns within email content, while Graph Neural Networks can learn from both node features and communication relationships to identify suspicious actors within a network. Community detection techniques further help in discovering groups of closely connected individuals that may exhibit unusual or coordinated behavior.

This project presents a comprehensive framework for detecting criminal communication networks within the Enron email dataset using Topic Modeling and Graph Neural Networks. The proposed methodology integrates BERTopic-based topic extraction, graph-based feature engineering, Node2Vec embeddings, centrality measures, and multiple Graph Neural Network architectures, including GCN, GraphSAGE, GAT, and TAGCN. Furthermore, Louvain Community Detection and Spectral Clustering are employed to identify suspicious communities and analyze the concentration of criminal actors within the communication network.

By combining text analytics, graph representation learning, and community analysis, the proposed approach aims to improve the identification of fraudulent communication patterns and provide deeper insights into the structure of criminal networks within organizational environments.

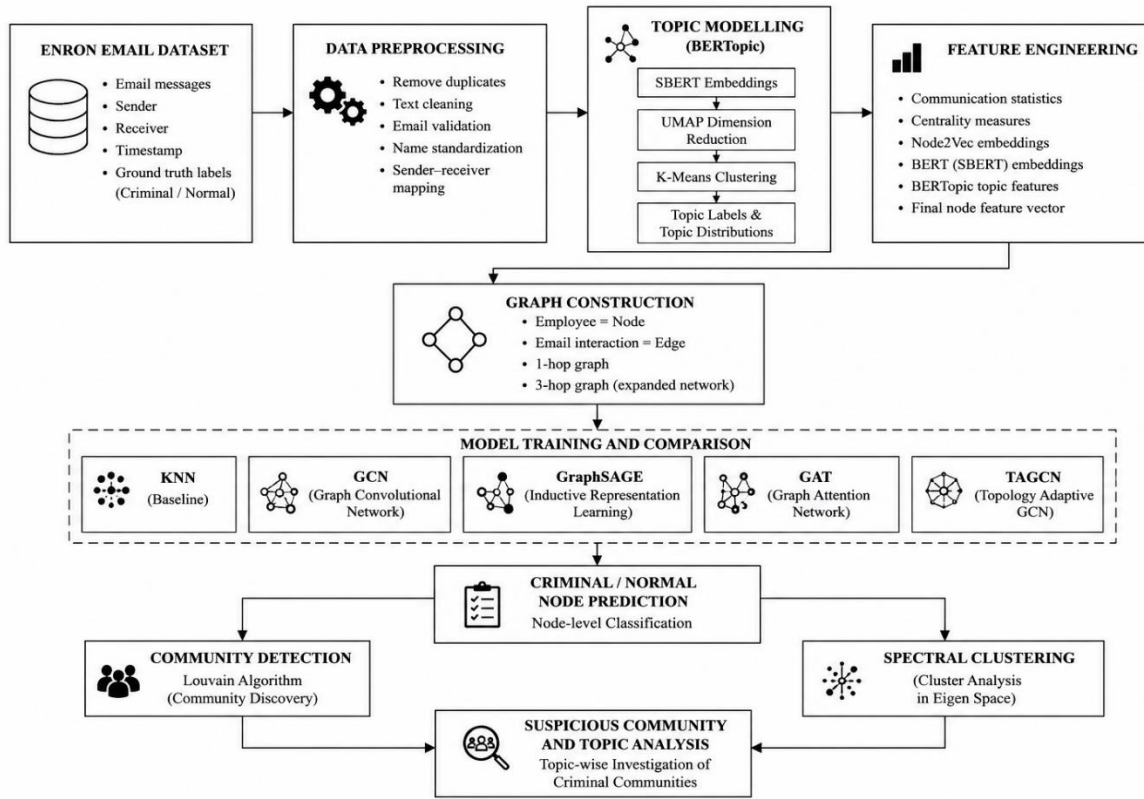


Fig 1.1-Overall Framework of the Proposed Criminal Communication Network Detection System

1.1 Background

Corporate organizations rely heavily on email communication for exchanging information, coordinating business operations, and making critical financial decisions. As organizations grow, millions of emails are generated between employees, creating complex communication networks. While the majority of these communications are legitimate, a small fraction may involve fraudulent activities, insider threats, financial manipulation, or unauthorized information exchange. Identifying such suspicious communication manually is extremely challenging due to the enormous volume of emails and the hidden relationships among individuals.

The Enron Corporation scandal, which emerged in 2001, remains one of the largest corporate fraud cases in history. Investigations revealed that fraudulent financial activities were coordinated through internal communications among company executives and employees. Following the company's collapse, more than half a million internal emails were released to the public, resulting in the Enron Email Dataset. This dataset has since become one of the most widely used benchmark datasets for research in fraud detection, email mining, social network analysis, and graph-based learning because it provides both the textual content of emails and the communication relationships between employees.

Traditional fraud detection approaches primarily focus on either analyzing the content of emails or examining communication patterns independently. However, criminal communication often depends on both what employees discuss and how they interact with one another. Consequently, analysing only textual information or only network structure may overlook important behavioural patterns associated with fraudulent activities.

Recent advances in Natural Language Processing (NLP), transformer-based language models, Graph Neural Networks (GNNs), and graph analytics have enabled more comprehensive analysis of large communication networks. Semantic topic modelling techniques such as BERTopic can automatically identify the major discussion topics within emails, while Graph Neural Networks can learn complex structural relationships among employees. By integrating semantic information with graph-based representations, it becomes possible to identify suspicious communication patterns more accurately and interpret the behaviour of criminal communication networks.

Motivated by these advancements, this project proposes an integrated framework that combines BERTopic-based topic modelling, graph representation learning, Graph Neural Networks, and community detection techniques to detect criminal communication networks within the Enron email dataset.

1.2 Motivation for the Study

Corporate fraud investigations often require analysts to examine thousands of email communications in an attempt to identify suspicious individuals and coordinated criminal activities. Manual investigation is time-consuming, expensive, and prone to human error, particularly when fraudulent communications are intentionally concealed within large volumes of legitimate emails. Traditional fraud detection techniques primarily rely on keyword matching, transactional analysis, or statistical methods, which frequently fail to capture the complex communication behaviour exhibited by individuals involved in coordinated fraudulent activities.

Although significant progress has been made in Natural Language Processing and Graph Neural Networks, many existing approaches analyse either the semantic content of emails or the structural relationships among employees independently. As a result, important behavioural patterns that arise from the combination of communication content and network interactions may remain undetected. Furthermore, community detection methods can identify groups of closely connected employees but do not explain the topics being discussed within those communities.

These limitations motivated the development of an integrated framework capable of analysing both the semantic and structural characteristics of organizational communication networks. By combining BERTopic-based topic modelling, graph representation learning, Graph Neural Networks, and community detection techniques, the proposed approach aims to identify suspicious individuals, detect criminal communication communities, and provide interpretable insights into the discussion topics associated with potentially fraudulent activities. Such a framework can support investigators in improving fraud detection, organizational risk assessment, and forensic analysis of large-scale communication networks.

1.3 Problem Statement

Detecting criminal communication networks within large-scale organizational email datasets is a challenging task because fraudulent communications are often concealed among vast numbers of legitimate emails and involve complex interactions between employees. Early identification of such communication patterns is important for supporting fraud investigations, identifying coordinated criminal behaviour, and reducing organizational risk. However, the highly imbalanced distribution of criminal and non-criminal instances further complicates the identification of suspicious individuals using conventional classification techniques.

Existing approaches primarily analyse either the semantic content of emails or the structural relationships within communication networks independently. Consequently, they are unable to fully capture the coordinated behaviour exhibited by employees involved in fraudulent activities. In addition, traditional community detection methods identify densely connected groups of employees but do not explain the dominant communication topics discussed within those communities, making it difficult for investigators to understand why a particular community may be considered suspicious.

Therefore, there is a need for an integrated framework that combines semantic topic modelling, graph representation learning, Graph Neural Networks, and community detection techniques to accurately identify criminal actors, detect suspicious communication communities, and interpret the dominant topics discussed within those communities in the Enron email network.

1.4 Aim and Objectives

Aim

To develop an intelligent and interpretable graph-based framework for detecting criminal communication networks in the Enron email dataset by integrating BERTopic-based semantic topic modelling, graph representation learning, Graph Neural Networks, and community detection techniques for accurate identification and analysis of suspicious individuals and communication communities.

Objectives

1. To preprocess and clean the Enron email dataset for reliable analysis.
2. To extract meaningful semantic topics from email communications using BERTopic.
3. To generate enriched employee-level node features by integrating communication statistics, centrality measures, Node2Vec embeddings, BERT embeddings, and topic-based features.
4. To construct 1-hop and 3-hop employee communication graphs representing interactions within the organization.

5. To evaluate and compare the performance of KNN, GCN, GraphSAGE, GAT, and TAGCN models for criminal node classification.
6. To identify suspicious communication communities using Louvain Community Detection and Spectral Clustering.
7. To analyse the dominant discussion topics within suspicious communities by integrating BERTopic with community detection results.
8. To evaluate the effectiveness of graph-based learning and semantic topic modelling for criminal communication network detection and fraud investigation.

1.5 Scope of the Project

This project focuses on analyzing the Enron email corpus to identify suspicious communication patterns and criminal actors within the organization's communication network. The study combines Natural Language Processing, Graph Analytics, and Graph Neural Networks to investigate both textual and structural aspects of employee communications.

The scope of the project includes email preprocessing, topic extraction, feature engineering, graph construction, criminal node classification, and community detection. The analysis is limited to the available Enron email dataset and the corresponding employee communication network derived from the dataset.

The proposed framework is intended for research and analytical purposes and can serve as a foundation for future fraud detection and insider threat monitoring systems in organizational environments.

1.6 Methodology Overview

The overall methodology adopted in this study is illustrated in Figure 1.1.

The proposed methodology consists of several interconnected stages designed to analyze the Enron email network comprehensively.

The process begins with data preprocessing, which includes cleaning email content, removing irrelevant information, handling duplicates, validating email addresses, and mapping employee identities. The cleaned emails are then processed using BERTopic to discover latent communication topics and generate topic-based features.

Next, employee-level features are created by combining communication statistics, centrality measures, Node2Vec embeddings, BERT embeddings, and topic-related information. These features are integrated into communication graphs where nodes represent employees and edges represent email interactions.

The generated graphs are used to train and evaluate multiple Graph Neural Network models, including GCN, GraphSAGE, GAT, and TAGCN. The best-performing model is selected based on classification performance metrics.

Finally, community detection techniques such as Louvain Community Detection and Spectral Clustering are applied to identify suspicious communication groups and analyze the distribution of criminal nodes within the network. The results are then evaluated to assess the effectiveness of the proposed framework in detecting criminal communication networks.

Chapter 2: Literature Review

2.1 Enron Fraud and Communication Network Analysis

The Enron Corporation scandal is widely recognized as one of the largest corporate fraud cases in history. Following the company's collapse in 2001, the Enron Email Corpus was publicly released and has since become one of the most widely used benchmark datasets for research in organizational communication analysis, fraud detection, insider threat identification, and social network analysis [1]–[4].

Several researchers have analyzed the Enron email corpus to investigate communication patterns, identify influential employees, and examine organizational structures using graph-based techniques. Early studies primarily focused on email frequency analysis, sender–receiver relationships, and network centrality measures such as Degree Centrality, Betweenness Centrality, and PageRank to identify important individuals within the communication network [3], [4]. Although these approaches successfully revealed structural communication patterns, they relied mainly on network topology and provided limited insight into the semantic content of email communications.

Subsequent research demonstrated that integrating textual analysis with communication networks provides a more comprehensive understanding of employee behaviour and organizational interactions. Such approaches enable investigators to analyze both how employees communicate and what they communicate about, thereby improving the identification of suspicious communication patterns and hidden criminal networks [3], [4].

2.2 Topic Modelling in Email Analysis

Topic modelling is a Natural Language Processing (NLP) technique used to automatically discover hidden semantic themes from large collections of text documents. It has been widely applied to email analysis, document clustering, and organizational communication studies to identify the major discussion topics present within textual data [5].

Early topic modelling research primarily relied on probabilistic methods such as Latent Dirichlet Allocation (LDA), which models each document as a mixture of latent topics represented by word probability distributions [5]. Although LDA has been successfully applied to organizational email datasets, its bag-of-words assumption limits its ability to capture contextual meaning and semantic relationships between words, often resulting in less coherent topics.

Recent advances in transformer-based language models have significantly improved semantic representation in topic modelling. BERT introduced contextual language embeddings capable of capturing word meaning based on surrounding context, while Sentence-BERT further enhanced

sentence-level semantic similarity by generating meaningful document embeddings [6], [7]. These embeddings are subsequently reduced using UMAP before clustering semantically similar documents into topics [8].

BERTopic combines transformer embeddings, dimensionality reduction, clustering, and class-based TF-IDF to generate highly coherent and interpretable topics [9]. Compared with traditional topic modelling techniques, BERTopic produces context-aware topics that are particularly suitable for analyzing complex organizational email communications. Consequently, BERTopic forms the foundation of the semantic analysis performed in this research.

2.3 Graph Neural Networks for Fraud Detection

Graph Neural Networks (GNNs) have emerged as one of the most effective deep learning techniques for analysing graph-structured data by simultaneously learning from node attributes and relationships between connected nodes [12], [16]. Unlike conventional machine learning algorithms that treat observations independently, GNNs exploit the communication structure of networks, making them highly suitable for fraud detection and organizational communication analysis.

Graph Convolutional Networks (GCN) introduced graph convolution for semi-supervised node classification by aggregating information from neighbouring nodes [12]. GraphSAGE extended this concept through inductive learning, enabling models to generalize to previously unseen nodes by learning neighbourhood aggregation functions [13]. Graph Attention Networks (GAT) further improved graph learning by introducing attention mechanisms that assign different importance weights to neighbouring nodes during feature aggregation [14].

Topology Adaptive Graph Convolutional Networks (TAGCN) addressed the limitations of conventional graph convolution by explicitly learning information from multiple graph hops while preserving the contribution of each neighbourhood distance [15]. This enables TAGCN to capture both local and long-range communication patterns, making it particularly effective for analysing organizational communication networks.

Recent studies have demonstrated that Graph Neural Networks significantly outperform traditional machine learning approaches for fraud detection because they effectively integrate graph topology with node-level features [17]–[19]. These capabilities make GNNs particularly suitable for identifying suspicious employees and detecting hidden criminal communication structures within the Enron email network.

2.4 Community Detection and Network Analysis

Community detection is a fundamental task in graph analysis that aims to identify groups of nodes that are more densely connected to one another than to the rest of the network [20], [22]. In

organizational communication networks, these communities often represent employees who communicate frequently, collaborate on common activities, or share similar behavioural patterns.

The Louvain Algorithm is one of the most widely adopted community detection techniques because of its computational efficiency and its ability to maximize graph modularity [20]. Numerous studies have successfully applied Louvain to large-scale social and communication networks for identifying densely connected communities and uncovering hidden organizational structures. However, Louvain relies primarily on graph connectivity and does not explain the semantic meaning of communications occurring within each community.

Spectral Clustering is another widely used graph clustering technique that partitions nodes by analysing the eigenvectors of graph similarity matrices [21]. Unlike modularity-based approaches, Spectral Clustering groups nodes according to structural similarity, making it effective for identifying behaviourally similar groups within complex networks. Nevertheless, similar to Louvain, it does not provide semantic interpretation of communication topics.

Recent research emphasizes integrating community detection with Natural Language Processing to improve the interpretability of network analysis [23]–[25]. Combining BERTopic with community detection enables investigators to identify not only which employees communicate closely, but also what they communicate about, thereby providing deeper insights into suspicious communication groups and improving fraud investigation.

2.5 Research Gap

The comparative analysis presented in **Table 2.1** highlights that although significant progress has been made in email analysis, fraud detection, and graph learning, several limitations remain in existing approaches. Many studies focus exclusively on either textual content or network structure, without effectively integrating both sources of information. Traditional machine learning methods often fail to capture complex communication relationships, while purely graph-based approaches may overlook important semantic information contained within email content.

Furthermore, limited research has explored the combined use of transformer-based topic modelling, graph representation learning, and community detection for criminal communication network identification in organizational email datasets. Existing approaches frequently analyse employee behaviour at the individual level but do not adequately investigate the formation of suspicious communication groups or interpret the dominant communication topics within those groups.

To address these limitations, this study proposes an integrated framework that combines BERTopic-based topic modelling, graph feature engineering, multiple Graph Neural Network architectures (GCN, GraphSAGE, GAT, and TAGCN), Louvain Community Detection, and Spectral Clustering to identify criminal actors, detect suspicious communication communities, and interpret the dominant communication topics discussed within those communities. The proposed framework jointly leverages communication content, graph structure, and community-

level semantic information to provide a more comprehensive and interpretable approach for criminal communication network detection.

Table 2.1: Comparative Analysis

Ref.	Author(s) & Year	Methodology	Advantages	Research Gap
[1]	Klimt & Yang (2004)	Enron Email Corpus	Introduced a benchmark dataset for organizational email analysis and fraud research.	Does not perform fraud detection or communication network analysis.
[3]	Diesner & Carley (2005)	Communication Network Analysis	Analysed employee interactions using graph-based communication networks.	Focuses mainly on network structure without semantic analysis of email content.
[5]	Blei et al. (2003)	Latent Dirichlet Allocation (LDA)	Classical probabilistic topic modelling technique for discovering latent topics.	Cannot capture contextual semantics and produces less coherent topics.
[9]	Grootendorst (2022)	BERTopic (SBERT + UMAP + c-TF-IDF)	Generates context-aware and highly interpretable topics using transformer embeddings.	Does not incorporate graph topology or communication network analysis.
[10]	Grover & Leskovec (2016)	Node2Vec	Learns continuous node embeddings while preserving network structure.	Does not utilize email content or node classification.
[12]	Kipf & Welling (2017)	Graph Convolutional Network (GCN)	Performs effective node classification using graph convolution.	Limited to fixed neighbourhood aggregation and lacks semantic topic information.
[13]	Hamilton et al. (2017)	GraphSAGE	Learns inductive node representations and generalizes to unseen nodes.	Does not integrate semantic communication features.

Ref.	Author(s) & Year	Methodology	Advantages	Research Gap
[14]	Veličković et al. (2018)	Graph Attention Network (GAT)	Uses attention mechanisms to assign different importance to neighbouring nodes.	Performance may degrade in highly imbalanced graph datasets.
[15]	Du et al. (2018)	Topology Adaptive Graph Convolutional Network (TAGCN)	Aggregates information from multiple graph hops while preserving hop-specific information.	Does not integrate topic modelling or community-level semantic analysis.
[20]	Blondel et al. (2008)	Louvain Community Detection	Efficiently detects densely connected communities by maximizing modularity.	Identifies communities based only on graph connectivity without explaining discussion topics.
[21]	von Luxburg (2007)	Spectral Clustering	Captures global graph structure and behavioural similarity among nodes.	Requires a predefined number of clusters and ignores semantic information.
[18]	Dou et al. (2020)	Graph Neural Network-based Fraud Detection	Demonstrates the effectiveness of GNNs for fraud detection in graph-structured data.	Does not integrate semantic topic modelling or community interpretation.

Chapter 3: System Design and Methodology

3.1 System Architecture Overview

The proposed framework integrates Natural Language Processing, Graph Analytics, Topic Modelling, and Graph Neural Networks to identify criminal communication networks within the Enron email dataset. The system begins with preprocessing and cleaning of email data, followed by topic extraction using BERTopic. Communication-based, structural, and semantic features are then generated for each employee and used to construct communication graphs. Multiple Graph Neural Network models are trained to classify employees as criminal or non-criminal. Finally, community detection techniques are applied to identify suspicious groups and validate criminal concentration within the communication network.

The overall workflow consists of Data Preprocessing, Feature Engineering, Topic Modelling, Graph Construction, Node Classification using GNNs, and Community Detection.

3.2 Dataset Description

This project uses three input files to construct the final communication network and identify criminal employees in the Enron email dataset.

The **main Enron email dataset** initially contained **447,417 email records with 32 attributes**, including sender information, receiver information, email content, metadata, and communication labels. This dataset served as the primary source for email preprocessing, topic modelling, feature extraction, and graph construction. After preprocessing, which included duplicate removal, email address normalization, invalid record filtering, recipient expansion, and text cleaning, the dataset was reduced to **327,696 valid email communications**, which were used for the subsequent stages of the proposed framework.

The **employee mapping dataset** contains the names of Enron employees together with their corresponding email addresses. Since many employees used multiple email addresses, this file was used to map different email IDs to a single employee identity, ensuring that each employee was represented as a unique node in the communication graph.

The **criminal employee label dataset** contains the list of **25 known criminal employees** identified during the Enron investigation. This file was used to assign node-level labels for the graph neural network models, where employees present in the list were labelled as criminal and all remaining employees were labelled as normal.

3.2.1 Main Enron Email Dataset

The main dataset contains the email communications exchanged among Enron employees. Each row represents an individual email along with sender, receiver, email body, metadata, and related

communication information. This file was used for text preprocessing, topic modelling, sender-receiver extraction, and graph construction.

Important columns include:

Column Description

From	Sender email address
To	Receiver email address
Subject	Subject of the email
Body	Original email content
X-From	Sender name from email metadata
X-To	Receiver name from email metadata
Date	Date of the email
Label	Email-level label indicating normal or suspicious/criminal communication

3.2.2 Employee Name and Email Mapping File

The second file contains employee names and their associated email addresses. Since the same employee may have multiple email IDs, this file was used to map email addresses to employee names and maintain consistent node identities in the graph.

Important columns include:

Column Description

num	Employee identifier
name	Employee name
email1	Primary email address
email2	Additional email address
email3	Additional email address
email4	Additional email address

3.2.3 Criminal Employee Label File

The third file contains the list of known criminal employees or Persons of Interest (POIs). This file was used to assign node-level labels for the graph-based classification task. Employees present in this file were labelled as criminal, while the remaining employees were treated as normal.

Important information included:

Field	Description
Employee Name	Name of the known criminal/POI employee
Label	Criminal employee indicator

These three files were combined during preprocessing to create the final employee communication network. The main email dataset provided the communication records, the mapping file ensured consistent employee identity resolution, and the criminal employee label file provided the ground-truth labels required for model training and evaluation. There was a total of 25 known criminals.

3.3 Data Preprocessing

The Enron email dataset initially contained **447,417 email records** collected over multiple years. As an initial preprocessing step, the temporal distribution of emails was analyzed to identify the period containing the most relevant communication data. The analysis showed that the majority of email exchanges occurred between **1999 and 2002**, as shown in **Figure 3.1**. However, to preserve sufficient historical context before and after the peak communication period, only emails exchanged between **1995 and 2003** were retained for further analysis, while records outside this period were discarded.

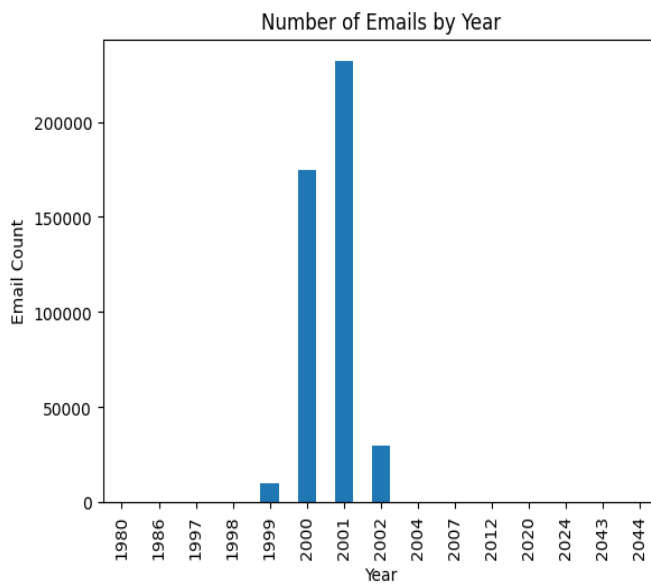


Fig 3.1-Distribution of Email Communications by Year

Following temporal filtering, several irrelevant metadata attributes, including Message-ID, MIME information, CC fields, BCC fields, mail identifiers, and other non-essential attributes, were removed to reduce data redundancy and improve processing efficiency.

The email bodies were then cleaned through a series of preprocessing steps, including:

- Conversion of text to lowercase.
- Removal of URLs and email addresses.
- Removal of punctuation and unnecessary symbols.
- Stopword removal.
- Removal of extra whitespace.
- Removal of forwarded and reply prefixes.
- Elimination of duplicate communications.
- Validation of sender and receiver email formats.
- Mapping multiple email addresses to unique employee names.

Finally, invalid records and emails with empty message bodies were discarded to ensure that only meaningful communications were retained for subsequent analysis.

After completing all preprocessing steps, the number of valid email communications was reduced from **447,417** to **327,696**, resulting in a cleaner and more reliable dataset for topic modelling, graph construction, feature engineering, and Graph Neural Network training.

3.4 Feature Engineering

Feature engineering was performed to capture communication behavior, network influence, and textual characteristics of employees.

The generated features included:

- Suspicious emails sent and received.
- Suspicious communication ratios.
- Degree Centrality.
- PageRank.
- Eigenvector Centrality.
- Betweenness Centrality.
- Node2Vec embeddings.
- BERT embeddings.
- Topic-based features derived from BERTopic.

These features provide a comprehensive representation of employee behavior from both structural and semantic perspectives.

3.5 Topic Modelling using BERTopic

BERTopic was employed to discover hidden communication themes within the Enron email corpus. The model utilizes transformer-based sentence embeddings to generate semantic representations of email content.

The topic modelling pipeline consisted of:

1. Generating sentence embeddings using the all-MiniLM-L6-v2 model.
2. Dimensionality reduction using UMAP.
3. Clustering semantically similar emails using K-Means clustering.
4. Extraction of representative keywords for each topic.

Ten communication topics were identified and manually interpreted based on their representative keywords and example emails. Topic distributions were subsequently used to generate employee-level topic features, including topic participation ratios and dominant communication topics.

3.6 Graph Construction

A communication graph was constructed from the processed email dataset.

In the graph:

- Nodes represent employees.
- Edges represent email interactions between employees.
- Edge weights represent communication frequency.

To capture broader communication patterns, multiple graph representations were generated, including 1-hop, 2-hop, and 3-hop communication graphs. These graphs enable the analysis of both direct and indirect employee interactions.

The graph structure forms the foundation for subsequent Graph Neural Network learning and community detection analysis.

3.7 Node Feature Generation

Each employee node was represented using a high-dimensional feature vector combining communication statistics, graph metrics, topic information, and semantic embeddings.

The final node feature representation consisted of:

- Communication-based features.
- Centrality measures.
- Node2Vec embeddings.
- BERT embedding features.
- Topic participation features.
- Dominant employee topic information.

This integrated representation allows Graph Neural Networks to learn from both employee behavior and communication content.

3.8 Graph Neural Network Models

Four Graph Neural Network architectures were evaluated for criminal node classification:

Graph Convolutional Network (GCN)

GCN aggregates information from neighboring nodes and learns node representations based on local graph structure.

GraphSAGE

GraphSAGE extends neighborhood aggregation by learning generalized aggregation functions that can effectively represent unseen nodes.

Graph Attention Network (GAT)

GAT introduces an attention mechanism that assigns different importance weights to neighboring nodes during feature aggregation.

Topology Adaptive Graph Convolutional Network (TAGCN)

TAGCN utilizes topology-adaptive filters to explicitly incorporate information from multiple graph hops. This enables the model to capture both local and global communication structures, making it particularly suitable for detecting hidden criminal relationships.

The performance of all models was evaluated using accuracy, precision, recall, F1-score, AUC, and Matthews Correlation Coefficient (MCC).

3.9 Community Detection using Louvain Algorithm

The Louvain Algorithm was applied to identify communities within the employee communication network. It partitions the graph by maximizing modularity, producing groups of employees that communicate more frequently with one another than with the rest of the network.

In this study, a **community** represents a group of employees with strong communication relationships. A **suspicious communication community** is defined as a community containing one or more employees labelled as criminal in the Enron dataset. Communities containing multiple criminal employees are considered more suspicious, as they may indicate coordinated communication patterns associated with fraudulent activities.

After detecting the communities, community-level analysis was performed by computing statistics such as community size, criminal count, and criminal ratio. BERTopic results were then integrated with the detected communities to identify the dominant discussion topics within each community. This combined structural and semantic analysis provides a more interpretable understanding of employee interactions and helps identify communication groups that may be associated with suspicious or fraudulent behaviour.

3.10 Spectral Clustering Analysis

Spectral Clustering was used as an additional community analysis technique to validate suspicious group formation within the network. Unlike modularity-based approaches, Spectral Clustering

utilizes node similarity information and feature representations to group employees with similar characteristics.

Multiple clustering configurations were evaluated, and the final clustering solution was selected based on modularity, criminal concentration, and cluster quality measures. Cluster-level analysis was performed to identify suspicious groups and examine their associated communication topics.

The results obtained from Spectral Clustering were compared with Louvain Community Detection to validate the consistency of suspicious employee group identification across different network analysis techniques.

Chapter 4: Implementation Details

4.1 Tools and Technologies Used

The project was implemented using Python and several open-source libraries for data preprocessing, natural language processing, graph analytics, and graph neural networks.

The primary development environment was Kaggle, which facilitated interactive data exploration and model development. Pandas and NumPy were used for data manipulation and preprocessing, while spaCy and NLTK were employed for text cleaning and tokenization. BERTopic, together with Sentence Transformers, UMAP, and Scikit-learn, was used for topic modelling. NetworkX was utilized for graph construction and computation of graph-based features, including centrality measures.

Node embeddings were generated using Node2Vec, and Graph Neural Network models were implemented using PyTorch and PyTorch Geometric. Community detection was performed using the Louvain algorithm, while Spectral Clustering was implemented using Scikit-learn.

4.2 Development Environment

The implementation was carried out using Python 3.x in a Kaggle environment. The experiments were executed on a Linux-based computing environment with sufficient computational resources to process the large-scale Enron email dataset.

Major libraries used include:

- Python
- Pandas
- NumPy
- NetworkX
- spaCy
- NLTK
- BERTopic
- Sentence Transformers
- UMAP
- Scikit-learn

- Node2Vec
- PyTorch
- PyTorch Geometric
- Matplotlib

Random seeds were fixed throughout the implementation to ensure reproducibility of experimental results.

4.3 Email Preprocessing Pipeline

The raw Enron email dataset contained multiple metadata fields, duplicate emails, inconsistent employee names, forwarded messages, and noisy textual content. A preprocessing pipeline was developed to improve data quality before analysis.

The implementation included:

- Removal of unnecessary metadata columns.
- Cleaning of forwarded and reply prefixes.
- Conversion of text to lowercase.
- Removal of URLs and email addresses.
- Removal of punctuation and stopwords.
- Elimination of duplicate email records.
- Validation of sender and receiver email addresses.
- Employee name mapping using reference datasets.
- Creation of a cleaned `Processed_Body` column.

The resulting dataset contained structured communication records suitable for topic modelling and graph construction.

4.4 BERTopic Implementation

Topic modelling was implemented using BERTopic to discover hidden semantic themes within employee communications.

Sentence embeddings were generated using the all-MiniLM-L6-v2 transformer model. UMAP was applied to reduce embedding dimensionality while preserving semantic relationships. K-Means clustering was then used to partition the reduced embeddings into ten clusters representing communication topics.

Each discovered topic was manually interpreted using representative keywords and sample emails. The dominant topic for every email was assigned and later incorporated into graph-based learning as an edge feature. Employee-level topic participation ratios and dominant node topics were also generated to enrich node representations.

Sentence embeddings were generated using the Sentence-BERT model, followed by dimensionality reduction using UMAP and topic clustering using the K-Means algorithm. The implementation details of these stages are presented in **Listings 4.1–4.3**.

```
from sentence_transformers import SentenceTransformer

embedding_model = SentenceTransformer(
    "all-MiniLM-L6-v2"
)

embeddings = embedding_model.encode(
    clean_texts,
    show_progress_bar=True,
    batch_size=64
)

print("Embeddings shape:", embeddings.shape)
```

Listing 4.1: Sentence-BERT Embedding Generation for Email Documents

```
from umap import UMAP

umap_model = UMAP(
    n_neighbors=10,
    n_components=5,
    min_dist=0.1,
    metric="cosine",
    low_memory=True,
    random_state=42
)
```

Listing 4.2: UMAP

```
from sklearn.cluster import KMeans

cluster_model = KMeans(
    n_clusters=10,
    random_state=42
)
```

Listing 4.3: K-Means Clustering Configuration for BERTopic

Configuration for BERTopic

Dimensionality Reduction

4.5 Graph Construction and Feature Integration

A communication graph was created where employees were represented as nodes and email interactions were represented as edges.

Graph-based features included:

- Communication statistics.
- Suspicious communication ratios.
- Degree Centrality.
- PageRank.
- Eigenvector Centrality.
- Betweenness Centrality.
- Node2Vec embeddings.
- BERT embedding features.
- BERTopic-derived topic features.

These heterogeneous features were integrated into a unified node feature matrix, providing both structural and semantic information for Graph Neural Network training.

4.6 K-Nearest Neighbors (KNN) Implementation

The K-Nearest Neighbors (KNN) algorithm was implemented as a baseline machine learning classifier to evaluate the effectiveness of graph-based learning methods. Unlike Graph Neural Networks, KNN does not utilize graph connectivity and relies solely on node feature vectors for classification.

The feature matrix consisted of communication statistics, graph centrality measures, Node2Vec embeddings, BERT embeddings, and BERTopic-derived topic features. The dataset was divided into training and testing sets using stratified sampling to preserve the class distribution.

For each test node, the algorithm identified the K nearest training samples based on feature similarity and assigned the class that occurred most frequently among its neighbors. Performance was evaluated using Accuracy, Precision, Recall, F1-score, AUC, and Matthews Correlation Coefficient (MCC).

The KNN model served as a benchmark for comparing the performance of graph-based deep learning models. Experimental results demonstrated that while KNN achieved high overall accuracy, its ability to exploit communication topology was limited compared with Graph Neural Network models, particularly TAGCN.

4.7 GCN Implementation

The Graph Convolutional Network (GCN) model was implemented using two graph convolution layers. The first layer transformed the high-dimensional node feature vectors into a lower-

dimensional hidden representation, while the second layer produced classification scores for criminal and non-criminal employees.

The model aggregates information from immediate neighboring nodes and learns node representations by combining graph topology with node attributes. Cross-entropy loss with class weighting was employed to address class imbalance during training.

4.8 GraphSAGE Implementation

GraphSAGE was implemented using neighborhood aggregation functions to learn node representations. Unlike GCN, GraphSAGE explicitly aggregates information from neighboring nodes using a learnable aggregation mechanism, enabling better generalization to unseen nodes.

The implementation consisted of two GraphSAGE convolution layers followed by nonlinear activation functions and supervised node classification.

4.9 GAT Implementation

Graph Attention Network (GAT) was implemented to investigate whether attention mechanisms could improve criminal node detection.

Multi-head attention was employed in the first graph attention layer to learn different communication patterns among neighboring employees. BERTopic-derived edge features were incorporated into the attention mechanism to provide semantic information about email topics during neighborhood aggregation.

Although GAT effectively learned attention weights, its performance was limited due to the highly imbalanced distribution of criminal nodes within the communication network.

4.10 TAGCN Implementation

Topology Adaptive Graph Convolutional Network (TAGCN) was implemented using three-hop graph information to capture both local and global communication patterns.

Unlike standard GCN, TAGCN explicitly models multiple hop neighborhoods within a single convolution operation, allowing it to distinguish information originating from one-hop, two-hop, and three-hop neighbors.

The enriched node features generated through communication statistics, graph embeddings, BERT embeddings, and BERTopic features were used as model input. TAGCN achieved the highest classification performance among all evaluated Graph Neural Network models.

4.11 Community Detection Implementation

Community detection was performed to identify groups of employees exhibiting dense communication patterns.

The Louvain algorithm was implemented to partition the communication graph into communities by maximizing modularity. Community-level statistics, including community size, criminal count, criminal ratio, and dominant communication topics, were computed to identify suspicious groups.

Spectral Clustering was subsequently applied to validate the detected communities using feature similarity. Cluster quality was evaluated using metrics such as Normalized Mutual Information (NMI), modularity, criminal concentration, and suspicious cluster count.

Finally, BERTopic results were integrated with both Louvain communities and Spectral Clusters to analyze the dominant communication topics within suspicious groups, providing additional semantic interpretation of criminal communication patterns.

Chapter 5: Results and Analysis

5.1 Dataset Statistics

After completing the preprocessing stage, the Enron email dataset was transformed into a structured communication dataset suitable for graph-based learning. Invalid email addresses, duplicate communications, empty email bodies, and unnecessary metadata were removed to improve data quality. Each employee was represented as a unique node, while email exchanges between employees formed the edges of the communication graph.

The processed dataset remained highly imbalanced, with only a small percentage of employees labeled as criminal. This imbalance created a challenging node classification problem and motivated the use of graph-based deep learning models with weighted loss functions. The final node feature matrix integrated communication statistics, graph centrality measures, Node2Vec embeddings, BERT embeddings, and BERTopic-derived topic features, providing a comprehensive representation of employee behavior.

5.2 Topic Modelling Results

BERTopic was successfully applied to identify semantic communication patterns within the Enron email corpus. Sentence embeddings generated using the all-MiniLM-L6-v2 transformer model were reduced using UMAP and clustered using K-Means to obtain ten communication topics.

Each topic was manually interpreted by examining representative keywords and sample emails. Topics included financial transactions, energy trading, investment research, credit risk monitoring, newsletters, and other business-related communications. The criminal rate for each topic was calculated by dividing the number of criminal emails within that topic by the total number of emails assigned to the topic. The topics generated by BERTopic, their manually assigned labels, and the corresponding criminal rates are presented in **Table 5.1**.

The analysis revealed that certain topics exhibited higher criminal communication rates than others, indicating that the semantic content of email communications can provide valuable information for fraud detection. Topic-based features generated from BERTopic were later incorporated into the Graph Neural Network models, improving the semantic representation of employee communications.

Table 5.1- BERTopic-Derived Topics with Manual Labels and Criminal Rates

Topic	Manual Label	Criminal Rate (%)
0	Agreements & Financial Transactions	0.12
1	Spam / Marketing / Newsletter	1.34
2	Financial News & Investment Research	0.54
3	Energy Markets / California Crisis	0.16
4	Energy Trading Transactions	0.23
5	Credit Risk Monitoring	0.44
6	Energy Policy News	0.05
7	Energy Market Manipulation Concerns	0.08
8	Stock Promotion Emails	0.02
9	Energy Contracts & Procurement	0.06

5.3 Graph Construction Statistics

The processed communication records were transformed into a graph where each employee represented a node and each email interaction represented an edge. Communication frequency was used to assign edge weights, capturing the strength of interactions between employees.

Multiple graph representations, including one-hop, two-hop, and three-hop communication graphs, were generated to investigate the effect of neighborhood information on node classification performance. For the final experiments, the three-hop communication graph was selected because it captures both direct and indirect communication relationships while preserving meaningful structural information.

Each node was represented using a high-dimensional feature vector consisting of communication statistics, graph centrality measures, Node2Vec embeddings, BERT embeddings, and BERTopic-derived features. The final graph served as the input for all Graph Neural Network models evaluated in this study.

5.4 K-Nearest Neighbors (KNN) Performance

The K-Nearest Neighbors (KNN) algorithm was implemented as a baseline machine learning classifier to compare its performance with Graph Neural Network models. Unlike graph-based models, KNN performs classification solely based on feature similarity without considering the communication structure between employees.

Experimental results showed that KNN achieved high overall accuracy because the majority of employees belonged to the normal class. However, its ability to identify criminal employees was limited due to the highly imbalanced nature of the dataset and the absence of graph structural information.

The performance of KNN established a baseline against which the effectiveness of Graph Neural Network models could be evaluated. The comparison demonstrated that incorporating graph topology significantly improved the identification of criminal communication patterns.

```
Confusion Matrix:
[[538  0]
 [ 4  1]]

Classification Report:
              precision    recall  f1-score   support

   Normal         0.99         1.00         1.00         538
   Criminal        1.00         0.20         0.33           5

 accuracy              0.99         0.99         0.99         543
 macro avg           1.00         0.60         0.66         543
 weighted avg        0.99         0.99         0.99         543

Additional Evaluation Metrics:
Accuracy: 0.992633517495396
Macro Precision: 0.996309963099631
Macro Recall: 0.6
Macro F1-score: 0.6648148148148149
AUC: 0.6
Matthews Correlation Coefficient (MCC): 0.4455603048296071
```

Figure 5.4.1: Performance Evaluation of KNN

5.5 Graph Neural Network Performance Comparison

Four Graph Neural Network architectures, namely Graph Convolutional Network (GCN), GraphSAGE, Graph Attention Network (GAT), and Topology Adaptive Graph Convolutional Network (TAGCN), were evaluated for criminal node classification.

Each model was trained using the same node feature matrix to ensure a fair comparison. Performance was evaluated using Accuracy, Precision, Recall, F1-score, Area Under the ROC Curve (AUC), and Matthews Correlation Coefficient (MCC).

Among the evaluated models, TAGCN achieved the best overall performance. Its ability to simultaneously aggregate information from one-hop, two-hop, and three-hop neighbors enabled it to capture both local and global communication patterns more effectively than the other models.

GraphSAGE also produced strong results by learning generalized neighborhood aggregation functions. GCN demonstrated competitive performance but was limited by its inability to distinguish information originating from different hop distances. GAT incorporated attention

mechanisms to assign different importance to neighboring nodes; however, its performance was affected by the severe class imbalance present in the dataset.

Overall, the experimental results demonstrate that integrating communication structure, semantic information from BERTopic, and topology-aware graph learning significantly improves criminal node classification.

GCN

```
Confusion Matrix:
[[431 107]
 [ 0   5]]

Classification Report:
              precision    recall  f1-score   support

   Normal         1.00        0.80        0.89         538
   Criminal        0.04        1.00        0.09           5

   accuracy              0.80         0.80         543
   macro avg           0.52        0.90        0.49         543
   weighted avg        0.99        0.80        0.88         543

Additional Evaluation Metrics:
Accuracy: 0.8029465930018416
Macro Precision: 0.5223214285714286
Macro Recall: 0.9005576208178439
Macro F1-score: 0.4875234844275092
AUC: 0.9628252788104089
Matthews Correlation Coefficient (MCC): 0.18911391616511855
```

Figure 5.5.1: Performance Evaluation of GCN

GraphSAGE

```
Confusion Matrix:
[[524  14]
 [ 0   5]]

Classification Report:
              precision    recall  f1-score   support

   Normal         1.00        0.97        0.99         538
   Criminal        0.26        1.00        0.42           5

   accuracy              0.97         0.97         543
   macro avg           0.63        0.99        0.70         543
   weighted avg        0.99        0.97        0.98         543

Additional Evaluation Metrics:
Accuracy: 0.9742173112338858
Macro Precision: 0.631578947368421
Macro Recall: 0.9869888475836431
Macro F1-score: 0.7017419962335216
AUC: 0.9992565055762083
Matthews Correlation Coefficient (MCC): 0.506270599364475
```

Figure 5.5.2: Performance Evaluation of GraphSAGE

Figure 5.4.3: Performance Evaluation of GAT

GAT				
Confusion Matrix:				
[[90 448]				
[0 5]]				
Classification Report:				
	precision	recall	f1-score	support
Normal	1.00	0.17	0.29	538
Criminal	0.01	1.00	0.02	5
accuracy			0.17	543
macro avg	0.51	0.58	0.15	543
weighted avg	0.99	0.17	0.28	543
Additional Evaluation Metrics:				
Accuracy: 0.17495395948434622				
Macro Precision: 0.5055187637969095				
Macro Recall: 0.5836431226765799				
Macro F1-score: 0.1542291324785136				
AUC: 0.4676579925650558				
Matthews Correlation Coefficient (MCC): 0.04297006573362292				

TAGCN				
Confusion Matrix:				
[[6361 12]				
[0 5]]				
Classification Report:				
	precision	recall	f1-score	support
Normal	1.00	1.00	1.00	6373
Criminal	0.29	1.00	0.45	5
accuracy			1.00	6378
macro avg	0.65	1.00	0.73	6378
weighted avg	1.00	1.00	1.00	6378
Additional Evaluation Metrics:				
Accuracy: 0.9981185324553151				
Macro Precision: 0.6470588235294118				
Macro Recall: 0.9990585281656991				
Macro F1-score: 0.7268015477533304				
AUC: 0.9997803232386632				
Matthews Correlation Coefficient (MCC): 0.5418153191793953				

Figure 5.5.4: Performance Evaluation of TAGCN

The performance evaluation results of the KNN, GCN, GraphSAGE, GAT, and TAGCN models, including the confusion matrices, classification reports, and additional evaluation metrics, are presented in Figures 5.4.1 and 5.5.1-5.5.4 respectively.

5.6 Community Detection Results

Community detection was performed using the Louvain Algorithm to identify densely connected groups of employees within the Enron communication network. The algorithm partitioned the graph into multiple communication communities by maximizing modularity, enabling the discovery of groups with strong internal communication patterns.

The analysis revealed that criminal employees were highly concentrated within a small number of communities rather than being uniformly distributed throughout the network. Among the detected communities, **Community 18** was identified as the most suspicious, containing **21 of the 25 known criminal employees**. This indicates that employees involved in fraudulent activities tended to communicate within closely connected groups, suggesting coordinated communication behaviour.

To further interpret these communities, BERTopic was integrated with the Louvain results. Each community was assigned a dominant discussion topic based on the most frequently occurring email topic within that community. The most suspicious community was primarily associated with **Topic 2 (Financial News and Investment Research)**, while substantial discussions related to **Topic 0 (Agreements and Financial Transactions)** and **Topic 1 (Spam, Marketing and Newsletter)** were also observed. This demonstrates that suspicious communities exhibit both strong structural connectivity and similar semantic communication patterns. The community-level statistics obtained using the Louvain algorithm are summarized in **Table 5.2**, including the size of each community, the number of criminal employees, and the dominant communication topic identified through BERTopic integration.

The results show that integrating community detection with topic modelling not only identifies densely connected groups of employees but also explains the dominant subjects discussed within those groups. Such semantic interpretation improves the explainability of detected suspicious communities and provides valuable insights for fraud investigation. Community 18 contained the highest number of criminal employees and was therefore selected for detailed analysis. The distribution of BERTopic topics within this community is presented in **Table 5.3**.

Table 5.2- Community-Level Summary Obtained Using the Louvian Algorithm

	Community	Community_Size	Criminal_Count	Dominant_Topic	Dominant_Topic_Email_Count
0	18	4093	21	2	7467
1	8	4082	0	0	24149
2	38	3152	1	0	10994
3	12	3129	0	1	8468
4	15	2506	0	0	8748
5	13	2242	1	1	4858
6	6	1816	0	2	4740
7	40	1459	2	0	7260
8	16	1376	0	1	4150
9	0	1292	0	3	1135
10	3	1213	0	0	3693
11	2	1138	0	0	5225
12	17	1121	0	0	2298
13	10	1117	0	1	1625
14	7	835	0	1	5018
15	19	390	0	1	1004
16	36	180	0	2	515
17	1	164	0	0	484
18	25	142	0	1	103
19	34	107	0	0	188

Table 5.3: BERTopic Distribution for the Most Suspicious Community Identified by the Louvain Algorithm

	Community	Topic	Topic_Email_Count
156	18.0	2	7467
154	18.0	0	6921
155	18.0	1	6256
157	18.0	3	1788
160	18.0	6	1675
158	18.0	4	1313
163	18.0	9	991
162	18.0	8	794
161	18.0	7	704
159	18.0	5	698

5.7 Spectral Clustering Results

Spectral Clustering was performed to further analyze the communication network and compare its clustering performance with the Louvain Algorithm. The communication graph was partitioned into **30 clusters**, and cluster quality was evaluated using **Normalized Mutual Information (NMI), modularity, criminal concentration, and suspicious cluster analysis**.

The results showed that **Cluster 16** was the most suspicious cluster, containing **22 of the 25 known criminal employees**, which represents a slightly higher criminal concentration than that achieved by the Louvain Algorithm. This indicates that Spectral Clustering was highly effective in grouping employees with similar communication characteristics, thereby concentrating the majority of criminal employees within a single cluster.

To improve cluster interpretability, BERTopic was integrated with the Spectral Clustering results. The dominant communication topic within each cluster was identified using the most frequently occurring email topic. The most suspicious cluster was primarily associated with **Topic 0 (Agreements and Financial Transactions)**, followed by **Topic 2 (Financial News and Investment Research)** and **Topic 1 (Spam, Marketing and Newsletter)**. This demonstrates that employees within the suspicious cluster not only shared similar communication patterns but also discussed closely related topics.

Although Spectral Clustering achieved a slightly lower **modularity score** than the Louvain Algorithm, it produced a higher concentration of criminal employees within a single cluster. This suggests that Spectral Clustering is particularly effective for grouping employees based on communication similarity, while Louvain provides stronger separation between communities. Together, both approaches offer complementary insights into the structure of criminal communication networks and improve the overall interpretability of fraud detection.

5.8 Discussion of Findings

The experimental results demonstrate the effectiveness of integrating Natural Language Processing, Graph Analytics, and Graph Neural Networks for criminal communication network detection.

BERTopic successfully extracted meaningful communication themes that enriched the node representations used during graph learning. The inclusion of semantic topic information improved the ability of Graph Neural Networks to distinguish criminal communication behavior from normal organizational communication.

Among all evaluated models, TAGCN consistently achieved the best classification performance by effectively utilizing multi-hop neighborhood information together with rich node features. The comparison with KNN further highlighted the importance of incorporating graph topology into fraud detection models.

Community detection using the Louvain Algorithm and Spectral Clustering provided complementary perspectives on employee interactions. While Louvain identified densely connected communication communities with higher modularity, Spectral Clustering demonstrated superior criminal concentration within individual clusters. The integration of BERTopic with both community detection approaches enabled meaningful interpretation of communication themes associated with suspicious employee groups.

Overall, the proposed framework successfully combines semantic analysis, graph representation learning, and community analysis to improve the detection and interpretation of criminal communication networks within the Enron email dataset.

The overall performance of the evaluated classification models is summarized in Table 5.4. The comparison includes Accuracy, Precision, Recall, F1-score, Matthews Correlation Coefficient

(MCC), and False Positives, providing a comprehensive evaluation of each model's ability to identify criminal employees within the Enron communication network.

Table 5.4: Comparative Performance of KNN, GCN, GraphSAGE, GAT, and TAGCN Models

Model	Accuracy (%)	Precision	Recall	F1-Score	False Positives	MCC
KNN	99.26	1.00	0.20	0.33	0	0.4456
GCN	80.29	0.04	1.00	0.09	107	0.1891
GraphSAGE	97.42	0.26	1.00	0.42	14	0.5063
GAT	17.50	0.01	1.00	0.02	448	0.0430
TAGCN	99.81	0.29	1.00	0.45	12	0.5418

The experimental results demonstrate that TAGCN integrated with BERTopic features achieved the best overall performance among the evaluated Graph Neural Network models. Although several models achieved high classification accuracy, accuracy alone is not a sufficient evaluation metric for this highly imbalanced dataset. Therefore, additional metrics such as precision, recall, false positives, and Matthews Correlation Coefficient (MCC) were considered.

TAGCN achieved the highest accuracy (99.81%), the highest MCC (0.5418), and perfect recall (1.00) while producing only 12 false positive predictions, indicating its ability to correctly identify criminal employees with very few misclassifications. Compared with GCN, GraphSAGE, and GAT, TAGCN provided a better balance between detecting criminal nodes and minimizing false alarms.

The improved performance can be attributed to the Topology Adaptive Graph Convolution (TAGConv), which effectively captures information from multiple communication hops while preserving the contributions of different neighborhood distances. Furthermore, integrating BERTopic-derived semantic features enabled the model to learn both the structural communication patterns and the dominant discussion topics associated with employees. This combination of graph topology and semantic information resulted in more discriminative node representations and improved criminal node classification.

Based on the overall evaluation, TAGCN with BERTopic features was selected as the final model for criminal communication network detection in the Enron email dataset.

Chapter 6: Conclusion And Future Work

6.1 Conclusion

This project presented a graph-based approach for detecting criminal communication networks within the Enron email corpus by integrating semantic topic modelling with Graph Neural Networks (GNNs). The proposed framework combined email content analysis, communication network analysis, and node-level structural features to identify suspicious individuals and communication patterns.

The raw Enron dataset underwent extensive preprocessing, including text cleaning, duplicate removal, email normalization, sender-receiver mapping, and construction of a communication graph. BERTopic was employed to discover latent discussion topics from email contents, and the resulting topic distributions were incorporated into node representations along with graph-based features such as centrality measures, communication statistics, and Node2Vec embeddings.

Multiple Graph Neural Network models, namely GCN, GraphSAGE, GAT, and TAGCN, were evaluated and compared with a traditional K-Nearest Neighbour (KNN) classifier. Experimental results demonstrated that incorporating BERTopic-derived semantic information significantly enhanced the performance of GraphSAGE, while TAGCN achieved the highest overall performance with an accuracy of **99.81%**, an **AUC of 0.9998**, and a **Macro F1-score of 0.7268**, successfully identifying all criminal nodes in the test set. Community analysis using the Louvain algorithm and Spectral Clustering further revealed densely connected communication groups and highlighted communities containing higher concentrations of criminal employees, providing valuable insights into organizational communication behaviour.

Overall, the proposed framework demonstrates that combining semantic topic modelling with graph representation learning provides an effective approach for analysing large-scale communication networks and identifying suspicious behavioural patterns. The integration of BERTopic, graph structural features, community detection techniques, and advanced GNN models offers a comprehensive methodology for criminal communication network analysis that can assist cybersecurity analysts and digital forensic investigators in prioritizing high-risk entities and understanding hidden communication structures.

6.2 Future Work

Although the proposed framework achieved promising results, several improvements can further enhance its applicability and performance.

The current study utilized BERTopic for semantic feature extraction. Future research can investigate more advanced Large Language Models (LLMs) and domain-specific transformer models to generate richer contextual embeddings for organizational communications.

Additional graph learning architectures, including Graph Transformers, Graph Attention Networks with improved attention mechanisms, and heterogeneous graph neural networks, can also be explored to model multiple types of entities and relationships present in enterprise communication networks.

The explainability of the proposed framework can be further improved by integrating Explainable Artificial Intelligence (XAI) techniques such as SHAP, LIME, and GNNExplainer. These approaches would help security analysts understand which node features, communication characteristics, and semantic topics contribute most to criminal predictions, thereby increasing model transparency and trustworthiness.

Future studies may also investigate overlapping community detection algorithms, hierarchical clustering techniques, and dynamic community evolution to better understand hidden organizational structures and collaborative criminal behaviour.

Finally, the proposed methodology can be validated on additional real-world communication datasets from different domains such as financial institutions, enterprise messaging systems, social media networks, and cybersecurity incident logs to evaluate its generalizability and robustness across diverse communication environments.

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